STM32 CubeMX

# 1. Description

## 1.1. Project

| | |
|---|---|
| Project Name | LwIP |
| Board Name | NUCLEO-H753ZI |
| Generated with: | STM32CubeMX 6.15.0 |
| Date | 01/09/2026 |

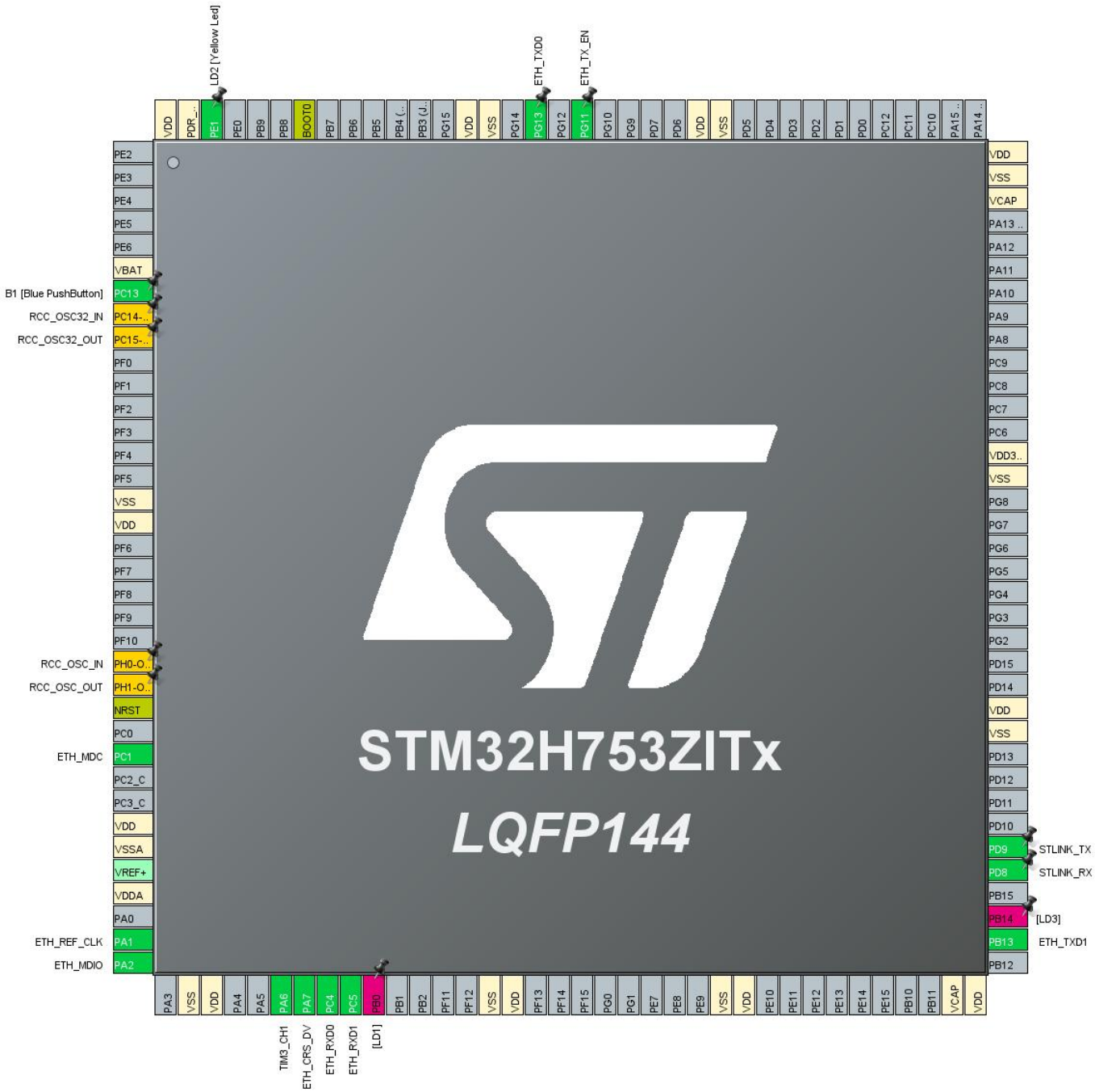
## 1.2. MCU

| | |
|---|---|
| MCU Series | STM32H7 |
| MCU Line | STM32H743/753 |
| MCU name | STM32H753ZITx |
| MCU Package | LQFP144 |
| MCU Pin number | 144 |

## 1.3. Core(s) information

| | |
|---|---|
| Core(s) | ARM Cortex-M7 |

## 2. Pinout Configuration



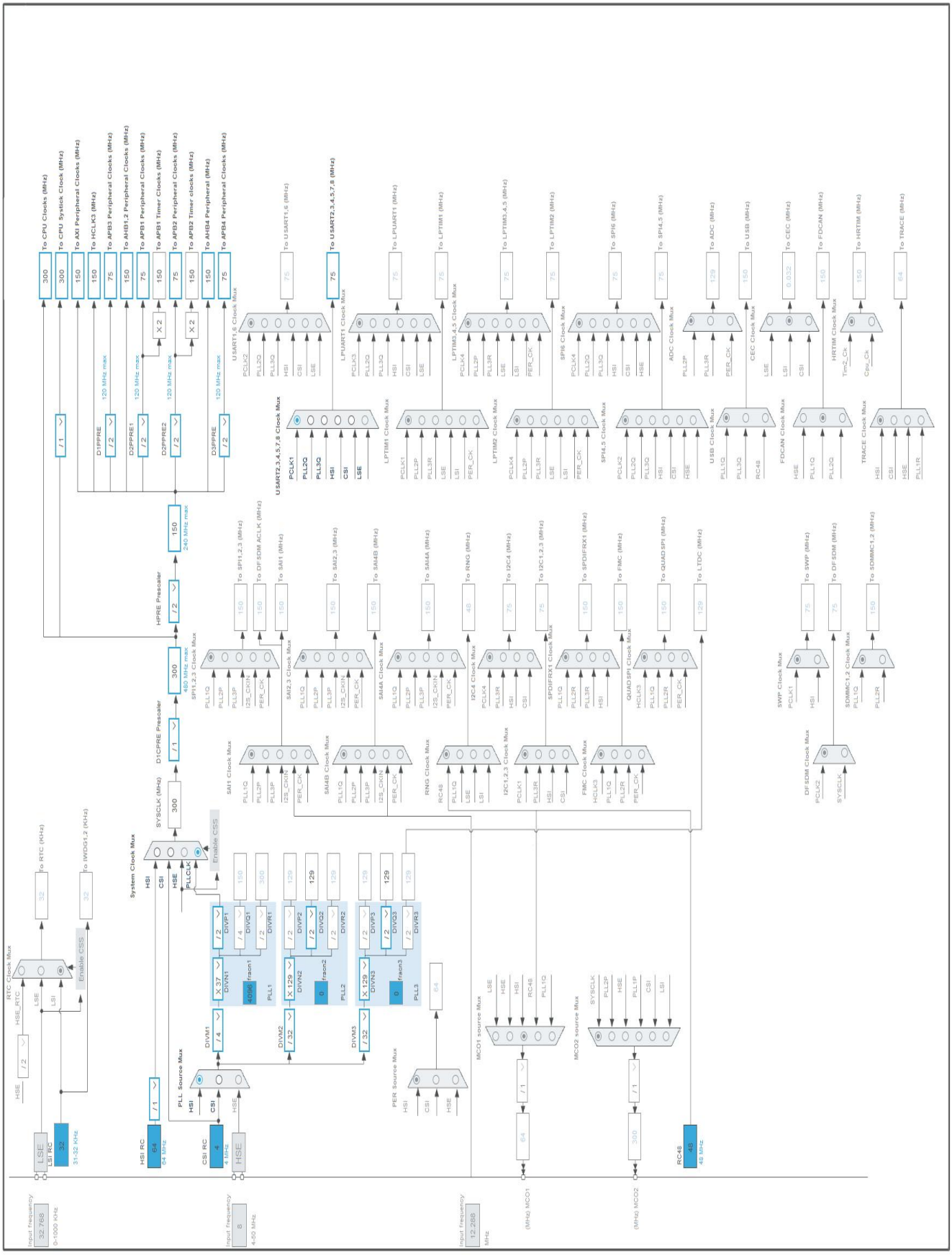
# 3. Pins Configuration

| Pin Number LQFP144 | Pin Name (function after reset) | Pin Type | Alternate Function(s) | Label |
|---|---|---|---|---|
| 6 | VBAT | Power | | |
| 7 | PC13 * | I/O | GPIO_Input | B1 [Blue PushButton] |
| 8 | PC14-OSC32_IN (OSC32_IN) ** | I/O | RCC_OSC32_IN | |
| 9 | PC15-OSC32_OUT (OSC32_OUT) ** | I/O | RCC_OSC32_OUT | |
| 16 | VSS | Power | | |
| 17 | VDD | Power | | |
| 23 | PH0-OSC_IN (PH0) ** | I/O | RCC_OSC_IN | |
| 24 | PH1-OSC_OUT (PH1) ** | I/O | RCC_OSC_OUT | |
| 25 | NRST | Reset | | |
| 27 | PC1 | I/O | ETH_MDC | |
| 30 | VDD | Power | | |
| 31 | VSSA | Power | | |
| 33 | VDDA | Power | | |
| 35 | PA1 | I/O | ETH_REF_CLK | |
| 36 | PA2 | I/O | ETH_MDIO | |
| 38 | VSS | Power | | |
| 39 | VDD | Power | | |
| 42 | PA6 | I/O | TIM3_CH1 | |
| 43 | PA7 | I/O | ETH_CRS_DV | |
| 44 | PC4 | I/O | ETH_RXD0 | |
| 45 | PC5 | I/O | ETH_RXD1 | |
| 46 | PB0 | I/O | | |
| 51 | VSS | Power | | |
| 52 | VDD | Power | | |
| 61 | VSS | Power | | |
| 62 | VDD | Power | | |
| 71 | VCAP | Power | | |
| 72 | VDD | Power | | |
| 74 | PB13 | I/O | ETH_TXD1 | |
| 75 | PB14 | I/O | | |
| 77 | PD8 | I/O | USART3_TX | STLINK_RX |
| 78 | PD9 | I/O | USART3_RX | STLINK_TX |
| 83 | VSS | Power | | |
| 84 | VDD | Power | | |

| Pin Number LQFP144 | Pin Name (function after reset) | Pin Type | Alternate Function(s) | Label |
|---|---|---|---|---|
| 94 | VSS | Power | | |
| 95 | VDD33_USB | Power | | |
| 106 | VCAP | Power | | |
| 107 | VSS | Power | | |
| 108 | VDD | Power | | |
| 120 | VSS | Power | | |
| 121 | VDD | Power | | |
| 126 | PG11 | I/O | ETH_TX_EN | |
| 128 | PG13 | I/O | ETH_TXD0 | |
| 130 | VSS | Power | | |
| 131 | VDD | Power | | |
| 138 | BOOT0 | Boot | | |
| 142 | PE1 * | I/O | GPIO_Output | LD2 [Yellow Led] |
| 143 | PDR_ON | Power | | |
| 144 | VDD | Power | | |

* The pin is affected with an I/O function
** The pin is affected with a peripheral function but no peripheral mode is activated

# 4. Clock Tree Configuration



# 1. Power Consumption Calculator report

## 1.1. Microcontroller Selection

| | |
|---|---|
| Series | STM32H7 |
| Line | STM32H743/753 |
| MCU | STM32H753ZITx |
| Datasheet | DS12117_Rev7 |

## 1.2. Parameter Selection

| | |
|---|---|
| Temperature | 25 |
| Vdd | 3.0 |

## 1.3. Battery Selection

| | |
|---|---|
| Battery | Alkaline(9V) |
| Capacity | 625.0 mAh |
| Self Discharge | 0.3 %/month |
| Nominal Voltage | 9.0 V |
| Max Cont Current | 200.0 mA |
| Max Pulse Current | 0.0 mA |
| Cells in series | 1 |
| Cells in parallel | 1 |

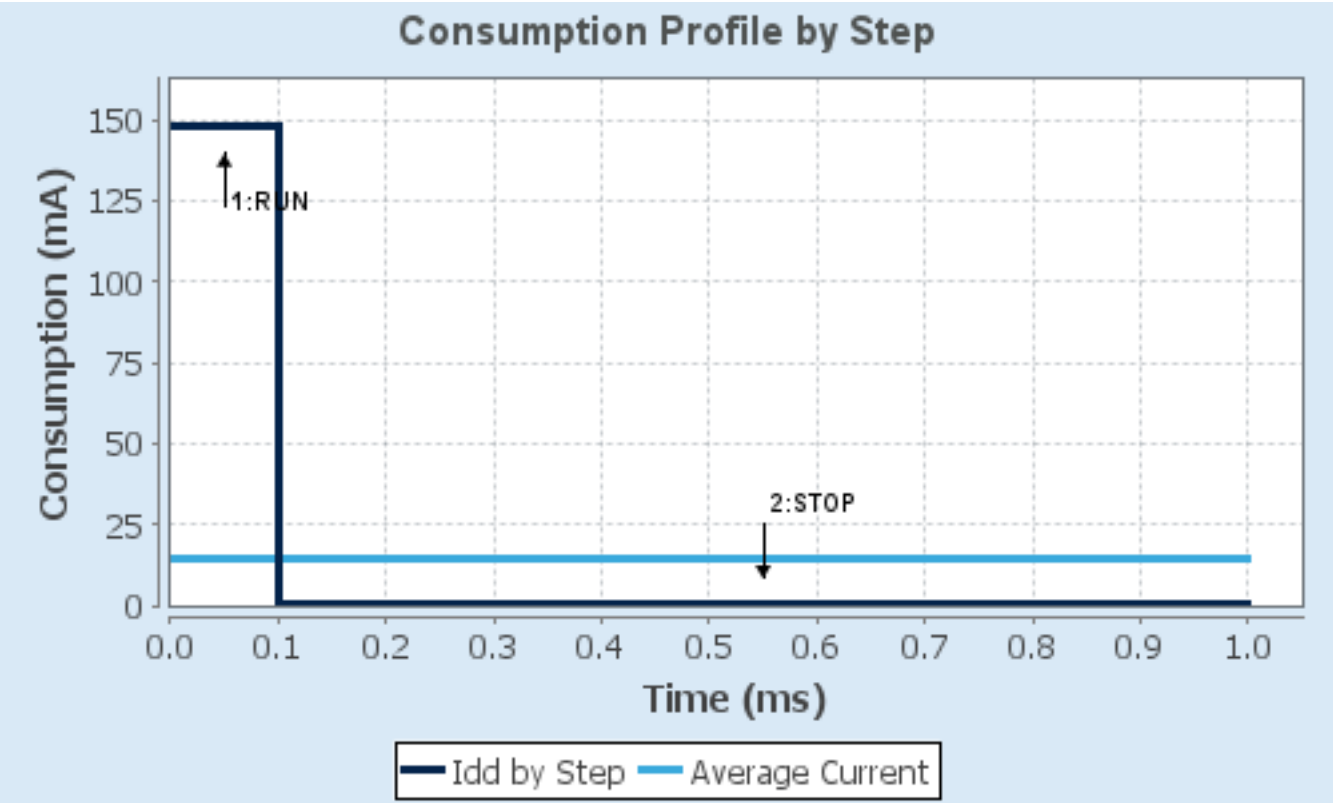
## 1.4. Sequence

| ***Step*** | Step1 | Step2 |
|---|---|---|
| ***Mode*** | RUN | STOP |
| ***Vdd*** | 3.0 | 3.0 |
| ***Voltage Source*** | Battery | Battery |
| ***Range*** | VOS0: Scale0-High | SVOS5: System-Scale5 |
| ***D1 Mode*** | DRUN/CRUN | DSTANDBY |
| ***D2 Mode*** | DRUN | DSTANDBY |
| ***D3 Mode*** | DRUN | DSTOP |
| ***Fetch Type*** | ITCM | NA |
| ***CPU Frequency*** | 480 MHz | 0 Hz |
| ***Clock Configuration*** | HSE BYP PLL | Flash-OFF |
| ***Clock Source Frequency*** | 24 MHz | 0 Hz |
| ***Peripherals*** | | |
| ***Additional Cons.*** | 0 mA | 0 mA |
| ***Average Current*** | 148 mA | 150 µA |
| ***Duration*** | 0.1 ms | 0.9 ms |
| ***DMIPS*** | 1027.0 | 0.0 |
| ***Ta Max*** | 105.46 | 124.98 |
| ***Category*** | In DS Table | In DS Table |

## 1.5. Results

| | | | |
|---|---|---|---|
| Sequence Time | 1 ms | Average Current | 14.94 mA |
| Battery Life | 1 day, 17 hours | Average DMIPS | 1027.2001 DMIPS |

## 1.6. Chart


Consumption Profile by Step
Consumption (mA)
150
125
100
75
50
25
0
1:RUN
2:STOP
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Time (ms)
Idd by Step
Average Current

# 2. Software Project

## 2.1. Project Settings

| Name | Value |
|---|---|
| Project Name | LwIP |
| Project Folder | C:\FREERTOS_LWIP_STM32H753ZI-master |
| Toolchain / IDE | STM32CubeIDE |
| Firmware Package Name and Version | STM32Cube FW_H7 V1.12.1 |
| Application Structure | Advanced |
| Generate Under Root | Yes |
| Do not generate the main() | No |
| Minimum Heap Size | 0x200 |
| Minimum Stack Size | 0x400 |

## 2.2. Code Generation Settings

| Name | Value |
|---|---|
| STM32Cube MCU packages and embedded software | Copy only the necessary library files |
| Generate peripheral initialization as a pair of '.c/.h' files | No |
| Backup previously generated files when re-generating | No |
| Keep User Code when re-generating | Yes |
| Delete previously generated files when not re-generated | Yes |
| Set all free pins as analog (to optimize the power consumption) | No |
| Enable Full Assert | No |

## 2.3. Advanced Settings - Generated Function Calls

| Rank | Function Name | Peripheral Instance Name |
|---|---|---|
| 1 | SystemClock_Config | RCC |
| 2 | MX_GPIO_Init | GPIO |
| 3 | MX_LWIP_Init | LWIP |
| 4 | MX_TIM3_Init | TIM3 |
| 5 | MX_USART3_UART_Init | USART3 |

# 3. Peripherals and Middlewares Configuration

## 3.1. CORTEX_M7

### 3.1.1. Parameter Settings:

**Speculation default mode Settings:**

| | |
|---|---|
| Speculation default mode | **Enabled *** |

**Cortex Interface Settings:**

| | |
|---|---|
| CPU ICache | **Enabled *** |
| CPU DCache | **Enabled *** |

**Cortex Memory Protection Unit Control Settings:**

| | |
|---|---|
| MPU Control Mode | Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers |

**Cortex Memory Protection Unit Region 0 Settings:**

| | |
|---|---|
| MPU Region | Enabled |
| MPU Region Base Address | **0x0 *** |
| MPU Region Size | 4GB |
| MPU SubRegion Disable | **0x87 *** |
| MPU TEX field level | level 0 |
| MPU Access Permission | ALL ACCESS NOT PERMITTED |
| MPU Instruction Access | DISABLE |
| MPU Shareability Permission | ENABLE |
| MPU Cacheable Permission | DISABLE |
| MPU Bufferable  Permission | DISABLE |

**Cortex Memory Protection Unit Region 1 Settings:**

| | |
|---|---|
| MPU Region | **Enabled *** |
| MPU Region Base Address | **0x30020000 *** |
| MPU Region Size | **128KB *** |
| MPU SubRegion Disable | **0x0 *** |
| MPU TEX field level | **level 1 *** |
| MPU Access Permission | **ALL ACCESS PERMITTED *** |
| MPU Instruction Access | **DISABLE *** |
| MPU Shareability Permission | DISABLE |
| MPU Cacheable Permission | DISABLE |
| MPU Bufferable  Permission | DISABLE |

**Cortex Memory Protection Unit Region 2 Settings:**

| | |
|---|---|
| MPU Region | **Enabled *** |
| MPU Region Base Address | **0x30040000 *** |
| MPU Region Size | **512B *** |

| | |
|---|---|
| MPU SubRegion Disable | **0x0 *** |
| MPU TEX field level | level 0 |
| MPU Access Permission | **ALL ACCESS PERMITTED *** |
| MPU Instruction Access | **DISABLE *** |
| MPU Shareability Permission | **ENABLE *** |
| MPU Cacheable Permission | DISABLE |
| MPU Bufferable  Permission | **ENABLE *** |

**Cortex Memory Protection Unit Region 3 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

**Cortex Memory Protection Unit Region 4 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

**Cortex Memory Protection Unit Region 5 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

**Cortex Memory Protection Unit Region 6 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

**Cortex Memory Protection Unit Region 7 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

**Cortex Memory Protection Unit Region 8 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

**Cortex Memory Protection Unit Region 9 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

**Cortex Memory Protection Unit Region 10 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

**Cortex Memory Protection Unit Region 11 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

**Cortex Memory Protection Unit Region 12 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

**Cortex Memory Protection Unit Region 13 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

**Cortex Memory Protection Unit Region 14 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

**Cortex Memory Protection Unit Region 15 Settings:**

| | |
|---|---|
| MPU Region | Disabled |

## 3.2. ETH

### Mode: RMII

### 3.2.1. Parameter Settings:

**General : Ethernet Configuration:**

| | |
|---|---|
| Warning | The ETH can work only when RAM is pointing at 0x24000000 |
| Note | PHY Driver must be configured from the LwIP 'Platform Settings' top right tab |
| Ethernet MAC Address | 00:80:E1:00:00:00 |
| Tx Descriptor Length | 4 |
| First Tx Descriptor Address | **0x30040200 *** |
| Rx Descriptor Length | 4 |
| First Rx Descriptor Address | **0x30040000 *** |
| Rx Buffers Address | **0x30000100 *** |
| Rx Buffers Length | 1536 |

## 3.3. MEMORYMAP
## mode: Activated

## 3.4. NUCLEO-H753ZI
## mode: Human Machine Interface

### 3.4.1. Human Machine Interface:

**Led:**

| | |
|---|---|
| USER LED GREEN (LD1) | **true *** |
| USER LED BLUE (LD2) | false |
| USER LED RED (LD3) | **true *** |

**Button:**

| | |
|---|---|
| USER BUTTON | Disable |

**VCOM:**

| | |
|---|---|
| Virtual Com Port | false |

**Demonstration code:**

| | |
|---|---|
| Generate demonstration code | Disabled |

## 3.5. RCC

### 3.5.1. Parameter Settings:

**Power Parameters:**

| | |
|---|---|
| SupplySource | PWR_LDO_SUPPLY |

| | |
|---|---|
| Power Regulator Voltage Scale | Power Regulator Voltage Scale 2 |

**RCC Parameters:**

| | |
|---|---|
| TIM Prescaler Selection | Disabled |
| HSE Startup Timout Value (ms) | 100 |
| LSE Startup Timout Value (ms) | 5000 |
| CSI Calibration Value | 32 |
| HSI Calibration Value | 64 |

**System Parameters:**

| | |
|---|---|
| VDD voltage (V) | 3.3 |
| Flash Latency(WS) | 2 WS (3 CPU cycle) |
| Product revision | rev.V |

**PLL range Parameters:**

| | |
|---|---|
| PLL1 clock Input range | Between 8 and 16 MHz |
| PLL1 clock Output range | Wide VCO range |

## 3.6. SYS

## Timebase Source: TIM6

## 3.7. TIM3

## Clock Source : Internal Clock

## Channel1: PWM Generation CH1

### 3.7.1. Parameter Settings:

**Counter Settings:**

| | |
|---|---|
| Prescaler (PSC - 16 bits value) | **50-1 *** |
| Counter Mode | Up |
| Counter Period (AutoReload Register - 16 bits value ) | **65535-1 *** |
| Internal Clock Division (CKD) | No Division |
| auto-reload preload | **Enable *** |

**Trigger Output (TRGO) Parameters:**

| | |
|---|---|
| Master/Slave Mode (MSM bit) | Disable (Trigger input effect not delayed) |
| Trigger Event Selection TRGO | Reset (UG bit from TIMx_EGR) |

**Clear Input:**

| | |
|---|---|
| Clear Input Source | Disable |

**PWM Generation Channel 1:**

| | |
|---|---|
| Mode | PWM mode 1 |
| Pulse (16 bits value) | **20 *** |
| Output compare preload | Enable |

| | |
|---|---|
| Fast Mode | Disable |
| CH Polarity | High |

## 3.8. USART3

## Mode: Asynchronous

### 3.8.1. Parameter Settings:

**Basic Parameters:**

| | |
|---|---|
| Baud Rate | 115200 |
| Word Length | 8 Bits (including Parity) |
| Parity | None |
| Stop Bits | 1 |

**Advanced Parameters:**

| | |
|---|---|
| Data Direction | Receive and Transmit |
| Over Sampling | 16 Samples |
| Single Sample | Disable |
| ClockPrescaler | 1 |
| Fifo Mode | Disable |
| Txfifo Threshold | 1 eighth full configuration |
| Rxfifo Threshold | 1 eighth full configuration |

**Advanced Features:**

| | |
|---|---|
| Auto Baudrate | Disable |
| TX Pin Active Level Inversion | Disable |
| RX Pin Active Level Inversion | Disable |
| Data Inversion | Disable |
| TX and RX Pins Swapping | Disable |
| Overrun | Enable |
| DMA on RX Error | Enable |
| MSB First | Disable |

## 3.9. FREERTOS

## Interface: CMSIS_V2

### 3.9.1. Config parameters:

**API:**

| | |
|---|---|
| FreeRTOS API | CMSIS v2 |

**Versions:**

| | |
|---|---|
| FreeRTOS version | 10.3.1 |

| | |
|---|---|
| CMSIS-RTOS version | 2.00 |

**MPU/FPU:**

| | |
|---|---|
| ENABLE_MPU | Disabled |
| ENABLE_FPU | Disabled |

**Kernel settings:**

| | |
|---|---|
| USE_PREEMPTION | Enabled |
| CPU_CLOCK_HZ | SystemCoreClock |
| TICK_RATE_HZ | 1000 |
| MAX_PRIORITIES | 56 |
| MINIMAL_STACK_SIZE | **512 *** |
| MAX_TASK_NAME_LEN | 16 |
| USE_16_BIT_TICKS | Disabled |
| IDLE_SHOULD_YIELD | Enabled |
| USE_MUTEXES | Enabled |
| USE_RECURSIVE_MUTEXES | Enabled |
| USE_COUNTING_SEMAPHORES | Enabled |
| QUEUE_REGISTRY_SIZE | 8 |
| USE_APPLICATION_TASK_TAG | Disabled |
| ENABLE_BACKWARD_COMPATIBILITY | Enabled |
| USE_PORT_OPTIMISED_TASK_SELECTION | Disabled |
| USE_TICKLESS_IDLE | Disabled |
| USE_TASK_NOTIFICATIONS | Enabled |
| RECORD_STACK_HIGH_ADDRESS | Disabled |

**Memory management settings:**

| | |
|---|---|
| Memory Allocation | Dynamic / Static |
| TOTAL_HEAP_SIZE | **30720 *** |
| Memory Management scheme | heap_4 |

**Hook function related definitions:**

| | |
|---|---|
| USE_IDLE_HOOK | Disabled |
| USE_TICK_HOOK | Disabled |
| USE_MALLOC_FAILED_HOOK | Disabled |
| USE_DAEMON_TASK_STARTUP_HOOK | Disabled |
| CHECK_FOR_STACK_OVERFLOW | **Option1 *** |

**Run time and task stats gathering related definitions:**

| | |
|---|---|
| GENERATE_RUN_TIME_STATS | Disabled |
| USE_TRACE_FACILITY | Enabled |
| USE_STATS_FORMATTING_FUNCTIONS | Disabled |

**Co-routine related definitions:**

| | |
|---|---|
| USE_CO_ROUTINES | Disabled |
| MAX_CO_ROUTINE_PRIORITIES | 2 |

**Software timer definitions:**

| | |
|---|---|
| USE_TIMERS | Enabled |
| TIMER_TASK_PRIORITY | 2 |
| TIMER_QUEUE_LENGTH | 10 |
| TIMER_TASK_STACK_DEPTH | 1024 |

**Interrupt nesting behaviour configuration:**

| | |
|---|---|
| LIBRARY_LOWEST_INTERRUPT_PRIORITY | 15 |
| LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY | 5 |

**Added with 10.2.1 support:**

| | |
|---|---|
| MESSAGE_BUFFER_LENGTH_TYPE | size_t |
| USE_POSIX_ERRNO | Disabled |

**CMSIS-RTOS V2 flags:**

| | |
|---|---|
| USE_OS2_THREAD_SUSPEND_RESUME | Enabled |
| USE_OS2_THREAD_ENUMERATE | Enabled |
| USE_OS2_EVENTFLAGS_FROM_ISR | Enabled |
| USE_OS2_THREAD_FLAGS | Enabled |
| USE_OS2_TIMER | Enabled |
| USE_OS2_MUTEX | Enabled |

## 3.9.2. Include parameters:

**Include definitions:**

| | |
|---|---|
| vTaskPrioritySet | Enabled |
| uxTaskPriorityGet | Enabled |
| vTaskDelete | Enabled |
| vTaskCleanUpResources | Disabled |
| vTaskSuspend | Enabled |
| vTaskDelayUntil | Enabled |
| vTaskDelay | Enabled |
| xTaskGetSchedulerState | Enabled |
| xTaskResumeFromISR | Enabled |
| xQueueGetMutexHolder | Enabled |
| xSemaphoreGetMutexHolder | Disabled |
| pcTaskGetTaskName | Disabled |
| uxTaskGetStackHighWaterMark | Enabled |
| xTaskGetCurrentTaskHandle | Enabled |
| eTaskGetState | Enabled |
| xEventGroupSetBitFromISR | Disabled |
| xTimerPendFunctionCall | Enabled |
| xTaskAbortDelay | Disabled |
| xTaskGetHandle | Disabled |
| uxTaskGetStackHighWaterMark2 | Disabled |

### 3.9.3. Advanced settings:

**Newlib settings (see parameter description first):**

| | |
|---|---|
| USE_NEWLIB_REENTRANT | **Enabled *** |

**Project settings (see parameter description first):**

| | |
|---|---|
| Use FW pack heap file | Enabled |

## 3.10. LWIP

## mode: Enabled

Advanced parameters are not listed except if modified by user.

### 3.10.1. General Settings:

**LwIP Version:**

| | |
|---|---|
| LwIP Version (Version of LwIP supported by CubeMX ** CubeMX specific **) | 2.1.2 |

**IPv4 - DHCP Options:**

| | |
|---|---|
| LWIP_DHCP (DHCP Module) | **Disabled *** |

**IP Address Settings:**

| | |
|---|---|
| IP_ADDRESS (IP Address) | **192.168.001.114 *** |
| NETMASK_ADDRESS (Netmask Address) | **255.255.255.000 *** |
| GATEWAY_ADDRESS (Gateway Address) | **192.168.001.001 *** |

**RTOS Dependency:**

| | |
|---|---|
| WITH_RTOS (Use FREERTOS ** CubeMX specific **) | Enabled |
| CMSIS_VERSION (CMSIS API Version used) | CMSIS v2 |
| RTOS_USE_NEWLIB_REENTRANT (RTOS used - 1) | Enabled |

**Platform Settings:**

| | |
|---|---|
| PHY Driver | Choose/LAN8742 |

**Protocols Options:**

| | |
|---|---|
| LWIP_ICMP (ICMP Module Activation) | Enabled |
| LWIP_IGMP (IGMP Module) | Disabled |
| LWIP_DNS (DNS Module) | Disabled |
| LWIP_UDP (UDP Module) | Enabled |
| MEMP_NUM_UDP_PCB (Number of UDP Connections) | 4 |
| LWIP_TCP (TCP Module) | Enabled |
| MEMP_NUM_TCP_PCB (Number of TCP Connections) | 5 |

## 3.10.2. Key Options:

| | |
|---|---|
| **Infrastructure - OS Awarness Option:** | |
| NO_SYS (OS Awarness) | OS Used |
| **Infrastructure - Timers Options:** | |
| LWIP_TIMERS (Use Support For sys_timeout) | Enabled |
| **Infrastructure - Core Locking and MPU Options:** | |
| SYS_LIGHTWEIGHT_PROT (Memory Functions Protection) | Enabled |
| **Infrastructure - Heap and Memory Pools Options:** | |
| MEM_SIZE (Heap Memory Size) | **131048 *** |
| LWIP_RAM_HEAP_POINTER (RAM Heap Pointer) | **0x30020000 *** |
| ETH_RX_BUFFER_CNT | 12 |
| **Infrastructure - Internal Memory Pool Sizes:** | |
| MEMP_NUM_PBUF (Number of Memory Pool struct Pbufs) | 16 |
| MEMP_NUM_RAW_PCB (Number of Raw Protocol Control Blocks) | 4 |
| MEMP_NUM_TCP_PCB_LISTEN (Number of Listening TCP Connections) | 8 |
| MEMP_NUM_TCP_SEG (Number of TCP Segments simultaneously queued) | 16 |
| MEMP_NUM_LOCALHOSTLIST (Number of Host Entries in the Local Host List) | 1 |
| **Pbuf Options:** | |
| PBUF_POOL_SIZE (Number of Buffers in the Pbuf Pool) | 16 |
| PBUF_POOL_BUFSIZE (Size of each pbuf in the pbuf pool) | 592 |
| **IPv4 - ARP Options:** | |
| LWIP_ARP (ARP Functionality) | Enabled |
| **Callback - TCP Options:** | |
| TCP_TTL (Number of Time-To-Live Used by TCP Packets) | 255 |
| TCP_WND (TCP Receive Window Maximum Size) | 5840 |
| TCP_QUEUE_OOSEQ (Allow Out-Of-Order Incoming Packets) | Enabled |
| LWIP_TCP_SACK_OUT (Allow Sending Selective Acknowledgements) | Disabled |
| TCP_MSS (Maximum Segment Size) | **1460 *** |
| TCP_SND_BUF (TCP Sender Buffer Space) | **5840 *** |
| TCP_SND_QUEUELEN (Number of Packet Buffers Allowed for TCP Sender) | **16 *** |
| **Network Interfaces Options:** | |
| LWIP_NETIF_STATUS_CALLBACK (Callback Function on Interface Status Changes) | Disabled |
| LWIP_NETIF_EXT_STATUS_CALLBACK (Extended Callback Function for several netif) | Disabled |
| LWIP_NETIF_LINK_CALLBACK (Callback Function on Interface Link Changes) | Enabled |
| **NETIF - Loopback Interface Options:** | |
| LWIP_NETIF_LOOPBACK (NETIF Loopback) | Disabled |
| **Infrastructure - Threading Options:** | |
| TCPIP_THREAD_NAME (TCPIP Thread Name) | "tcpip_thread" |

| | |
|---|---|
| TCPIP_THREAD_STACKSIZE (TCPIP Thread Stack Size) | **2048 *** |
| TCPIP_THREAD_PRIO (TCPIP Thread Priority Level) | 24 |
| TCPIP_MBOX_SIZE (TCPIP Mailbox Size) | 6 |
| DEFAULT_THREAD_NAME (Default LwIP Thread Name) | "lwIP" |
| DEFAULT_THREAD_STACKSIZE (Default LwIP Thread Stack Size) | **2048 *** |
| DEFAULT_THREAD_PRIO (Default LwIP Thread Priority Level) | 3 |
| DEFAULT_RAW_RECVMBOX_SIZE (Default Mailbox Size on a NETCONN Raw) | 0 |
| DEFAULT_TCP_RECVMBOX_SIZE (Default Mailbox Size on a NETCONN TCP) | 6 |
| DEFAULT_ACCEPTMBOX_SIZE (Default Mailbox Size for Incoming Connections) | 6 |

**Thread Safe APIs - Netconn Options:**

| | |
|---|---|
| LWIP_NETCONN (NETCONN API) | Enabled |

**Thread Safe APIs - Socket Options:**

| | |
|---|---|
| LWIP_SOCKET (Socket API) | Enabled |
| LWIP_COMPAT_SOCKETS (BSD-style Socket Functions Names) | 1 |
| LWIP_SOCKET_OFFSET (Socket Offset Number) | 0 |
| LWIP_SOCKET_SELECT (Select for Socket) | Enabled |
| LWIP_SOCKET_POLL (Poll for Socket) | Enabled |

## 3.10.3. PPP:

**PPP Options:**

| | |
|---|---|
| PPP_SUPPORT (PPP Module) | Disabled |

## 3.10.4. IPv6:

**IPv6 Options:**

| | |
|---|---|
| LWIP_IPV6 (IPv6 Protocol) | Disabled |

## 3.10.5. HTTPD:

**HTTPD Options:**

| | |
|---|---|
| LWIP_HTTPD (LwIP HTTPD Support ** CubeMX specific **) | Disabled |

## 3.10.6. SNMP:

**SNMP Options:**

| | |
|---|---|
| LWIP_SNMP (LwIP SNMP Agent) | Disabled |

## 3.10.7. SNTP/SMTP:

**SNTP Options:**

| | |
|---|---|
| LWIP_SNTP (LWIP SNTP Support ** CubeMX specific **) | Disabled |

**SMTP Options:**

| | |
|---|---|
| LWIP_SMTP (LWIP SMTP Support ** CubeMX specific **) | Disabled |

## 3.10.8. MDNS/TFTP:

**MDNS Options:**

| | |
|---|---|
| LWIP_MDNS (Multicast DNS Support ** CubeMX specific **) | Disabled |

**TFTP Options:**

| | |
|---|---|
| LWIP_TFTP (TFTP Support ** CubeMX specific **) | Disabled |

## 3.10.9. Perf/Checks:

**Sanity Checks:**

| | |
|---|---|
| LWIP_DISABLE_TCP_SANITY_CHECKS (TCP Sanity Checks) | Disabled |
| LWIP_DISABLE_MEMP_SANITY_CHECKS (MEMP Sanity Checks) | Disabled |

**Performance Options:**

| | |
|---|---|
| LWIP_PERF (Performace Testing for LwIP) | Disabled |

## 3.10.10. Statistics:

**Debug - Statistics Options:**

| | |
|---|---|
| LWIP_STATS (Statictics Collection) | Disabled |

## 3.10.11. Checksum:

**Infrastructure - Checksum Options:**

| | |
|---|---|
| CHECKSUM_BY_HARDWARE (Hardware Checksum ** CubeMX specific **) | Enabled |
| LWIP_CHECKSUM_CTRL_PER_NETIF (Generate/Check Checksum per Netif) | Disabled |
| CHECKSUM_GEN_IP (Generate Software Checksum for Outgoing IP Packets) | Disabled |
| CHECKSUM_GEN_UDP (Generate Software Checksum for Outgoing UDP Packets) | Disabled |
| CHECKSUM_GEN_TCP (Generate Software Checksum for Outgoing TCP Packets) | Disabled |
| CHECKSUM_GEN_ICMP (Generate Software Checksum for Outgoing ICMP Packets) | Enabled |
| CHECKSUM_GEN_ICMP6 (Generate Software Checksum for Outgoing ICMP6 Packets) | Disabled |
| CHECKSUM_CHECK_IP (Generate Software Checksum for Incoming IP Packets) | Disabled |
| CHECKSUM_CHECK_UDP (Generate Software Checksum for Incoming UDP Packets) | Disabled |

| | |
|---|---|
| CHECKSUM_CHECK_TCP (Generate Software Checksum for Incoming TCP Packets) | Disabled |
| CHECKSUM_CHECK_ICMP (Generate Software Checksum for Incoming ICMP Packets) | Enabled |
| CHECKSUM_CHECK_ICMP6 (Generate Software Checksum for Incoming ICMP6 Packets) | Disabled |

## 3.10.12. Debug:

**LwIP Main Debugging Options:**

| | |
|---|---|
| LWIP_DBG_MIN_LEVEL (Minimum Level) | All |

## 3.10.13. Platform Settings:

| | |
|---|---|
| Driver_PHY | LAN8742 |

*** User modified value**

# 4. System Configuration

## 4.1. GPIO configuration

| IP | Pin | Signal | GPIO mode | GPIO pull/up pull down | Max Speed | User Label |
|---|---|---|---|---|---|---|
| ETH | PC1 | ETH_MDC | Alternate Function Push Pull | No pull-up and no pull-down | **Very High *** | |
| | PA1 | ETH_REF_CLK | Alternate Function Push Pull | No pull-up and no pull-down | **Very High *** | |
| | PA2 | ETH_MDIO | Alternate Function Push Pull | No pull-up and no pull-down | **Very High *** | |
| | PA7 | ETH_CRS_DV | Alternate Function Push Pull | No pull-up and no pull-down | **Very High *** | |
| | PC4 | ETH_RXD0 | Alternate Function Push Pull | No pull-up and no pull-down | **Very High *** | |
| | PC5 | ETH_RXD1 | Alternate Function Push Pull | No pull-up and no pull-down | **Very High *** | |
| | PB13 | ETH_TXD1 | Alternate Function Push Pull | No pull-up and no pull-down | **Very High *** | |
| | PG11 | ETH_TX_EN | Alternate Function Push Pull | No pull-up and no pull-down | **Very High *** | |
| | PG13 | ETH_TXD0 | Alternate Function Push Pull | No pull-up and no pull-down | **Very High *** | |
| TIM3 | PA6 | TIM3_CH1 | Alternate Function Push Pull | No pull-up and no pull-down | Low | |
| USART3 | PD8 | USART3_TX | Alternate Function Push Pull | No pull-up and no pull-down | Low | STLINK_RX |
| | PD9 | USART3_RX | Alternate Function Push Pull | No pull-up and no pull-down | Low | STLINK_TX |
| Single Mapped Signals | PC14-OSC32_IN (OSC32_IN) | RCC_OSC32_IN | n/a | n/a | n/a | |
| | PC15-OSC32_OUT | RCC_OSC32_OUT | n/a | n/a | n/a | |
| | PH0-OSC_IN (PH0) | RCC_OSC_IN | n/a | n/a | n/a | |
| | PH1-OSC_OUT (PH1) | RCC_OSC_OUT | n/a | n/a | n/a | |
| GPIO | PC13 | GPIO_Input | Input mode | No pull-up and no pull-down | n/a | B1 [Blue PushButton] |
| | PE1 | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low | LD2 [Yellow Led] |

### 4.2. DMA configuration

nothing configured in DMA service

### 4.3. BDMA configuration

nothing configured in DMA service

### 4.4. MDMA configuration

nothing configured in DMA service

## 4.5. NVIC configuration

### 4.5.1. NVIC

| Interrupt Table | Enable | Preenmption Priority | SubPriority |
|---|---|---|---|
| Non maskable interrupt | true | 0 | 0 |
| Hard fault interrupt | true | 0 | 0 |
| Memory management fault | true | 0 | 0 |
| Pre-fetch fault, memory access fault | true | 0 | 0 |
| Undefined instruction or illegal state | true | 0 | 0 |
| System service call via SWI instruction | true | 0 | 0 |
| Debug monitor | true | 0 | 0 |
| Pendable request for system service | true | 15 | 0 |
| System tick timer | true | 15 | 0 |
| TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts | true | 15 | 0 |
| Ethernet global interrupt | true | 5 | 0 |
| PVD and AVD interrupts through EXTI line 16 | unused | | |
| Flash global interrupt | unused | | |
| RCC global interrupt | unused | | |
| TIM3 global interrupt | unused | | |
| USART3 global interrupt | unused | | |
| Ethernet wake-up interrupt through EXTI line 86 | unused | | |
| FPU global interrupt | unused | | |
| HSEM1 global interrupt | unused | | |

### 4.5.2. NVIC Code generation

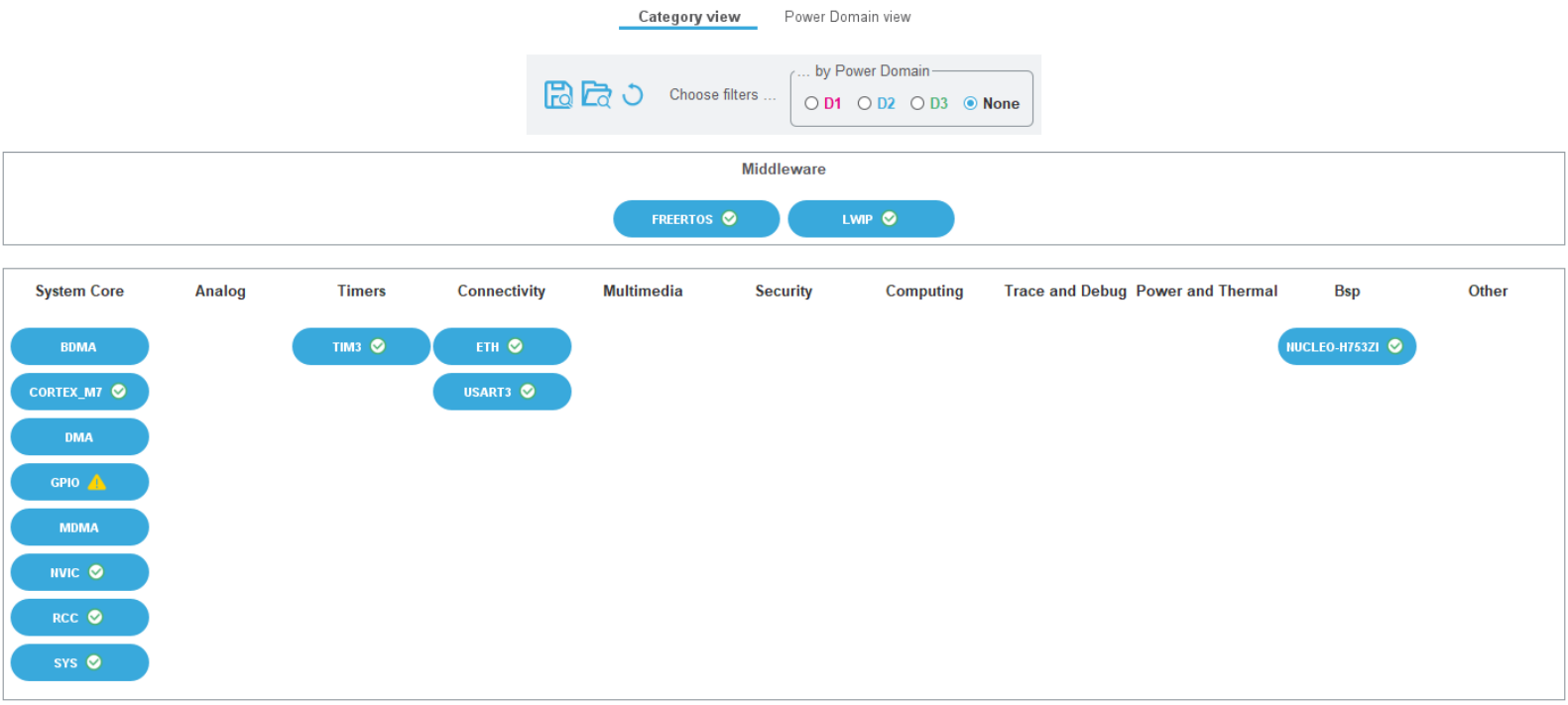
| Enabled interrupt Table | Select for init sequence ordering | Generate IRQ handler | Call HAL handler |
|---|---|---|---|
| Non maskable interrupt | false | true | false |
| Hard fault interrupt | false | true | false |
| Memory management fault | false | true | false |
| Pre-fetch fault, memory access fault | false | true | false |
| Undefined instruction or illegal state | false | true | false |
| System service call via SWI instruction | false | false | false |
| Debug monitor | false | true | false |
| Pendable request for system service | false | false | false |
| System tick timer | false | false | true |
| TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts | false | true | true |
| Ethernet global interrupt | false | true | true |

*** User modified value**

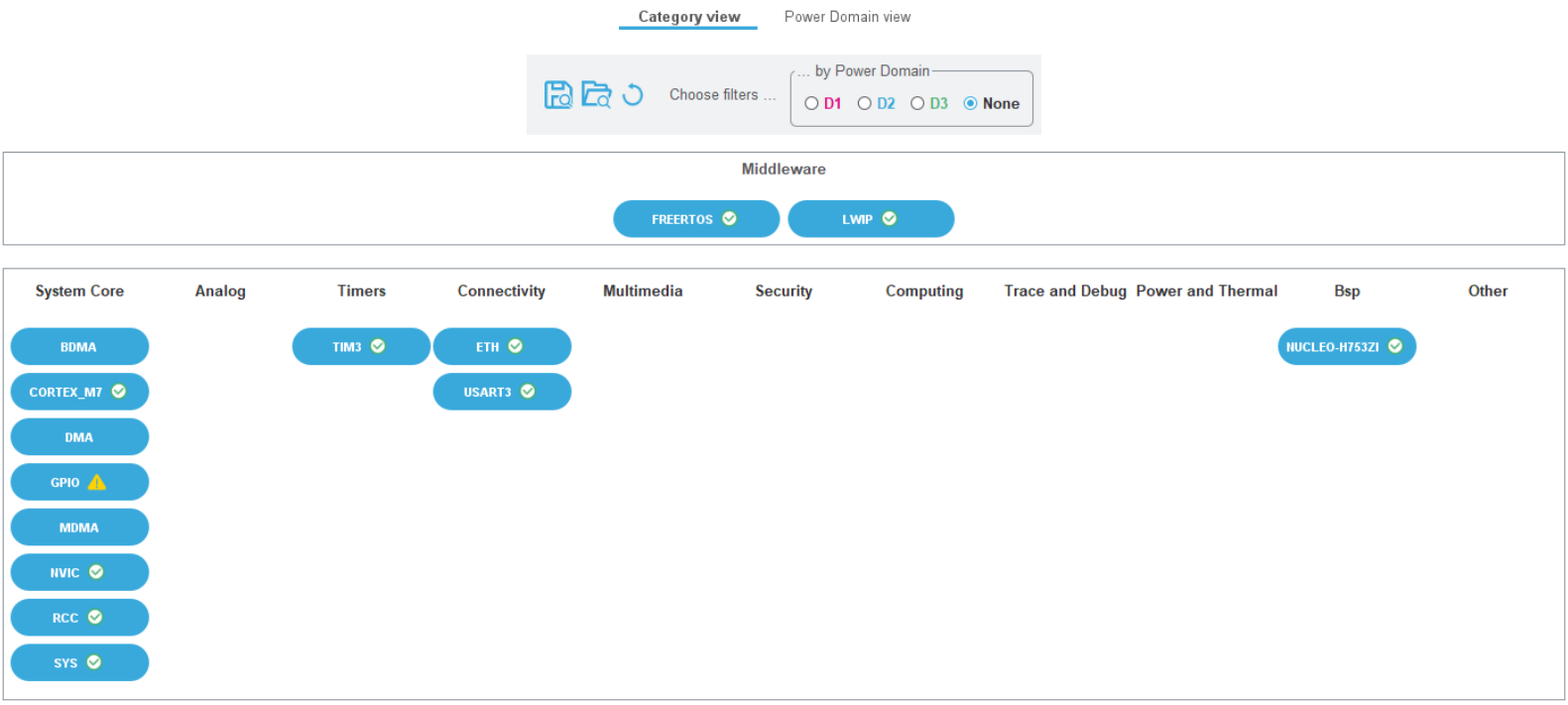
# 5. System Views

## 5.1. Category view

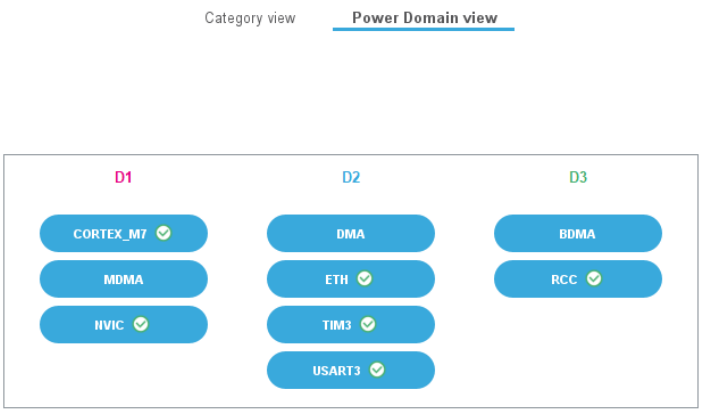
### *5.1.1. Current*



### *5.1.2. Without filters*



## 5.2. Power Domain view



# 6. Docs & Resources

| Type | Link |
|---|---|
| BSDL files | https://www.st.com/resource/en/bsdl_model/stm32h7_bsdl.zip |
| IBIS models | https://www.st.com/resource/en/ibis_model/stm32h7_ibis.zip |
| System View Description | https://www.st.com/resource/en/svd/stm32h7-svd.zip |
| Presentations | https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h7_series_product_overview.pdf |
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| Flyers | https://www.st.com/resource/en/flyer/flpowerstbd.pdf |
| Flyers | https://www.st.com/resource/en/flyer/flstm32h7rs.pdf |
| Security Bulletin | https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf |
| Security Bulletin | https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf |

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| Application Notes | https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4539-hrtim-cookbook-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4839-level-1-cache-on-stm32f7-series-and-stm32h7-series-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4891-stm32h72x-stm32h73x-and-singlecore-stm32h74x75x-system-architecture-and-performance-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4936-migration-of-microcontroller-applications-from-stm32f7-series-to-stm32h743753-line-stmicroelectronics.pdf |

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| Application Notes | https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4990-getting-started-with-sigmadelta-digital-interface-on-applicable-stm32-microcontrollers-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf |
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| Application Notes | https://www.st.com/resource/en/application_note/an5312-migration-from-revy-to-revv-for-stm32h743753-and-stm32h750-value-line-microcontrollers-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an5354-getting-started-with-the-stm32h7-series-mcu-16bit-adc-stmicroelectronics.pdf |
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| Application Notes | https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an5293-migration-guide-from-stm32f7-series-to-stmh74x75x-stm32h72x73x-and-stmh7a37bx-devices-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf |

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| Application Notes | https://www.st.com/resource/en/application_note/an5927-i3c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf |
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| Application Notes | https://www.st.com/resource/en/application_note/an4566-how-to-extend-the-dac-performance-on-stm32-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an5647-the-goertzel-algorithm-to-compute-individual-terms-of-the-discrete-fourier-transform-dft-in-stm32-products-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an2606-introduction-to-system-memory-boot-mode-on-stm32-mcus-stmicroelectronics.pdf |
| Application Notes | https://www.st.com/resource/en/application_note/an5200-getting-started-with-the-stm32h7-mcu-sdmmc-host-controller-stmicroelectronics.pdf |
| Application Notes for related Tools & Software | https://www.st.com/resource/en/application_note/an4323-getting-started-with-stemwin-library-stmicroelectronics.pdf |
| Application Notes for related Tools & Software | https://www.st.com/resource/en/application_note/an4435-guidelines-for-obtaining-ulcsaiec-607301603351-class-b-certification-in-any-stm32-application-stmicroelectronics.pdf |
| Application Notes for related Tools & Software | https://www.st.com/resource/en/application_note/an4657-stm32-inapplication-programming-iap-using-the-usart-stmicroelectronics.pdf |
| Application Notes for related Tools & Software | https://www.st.com/resource/en/application_note/an4841-digital-signal-processing-for-stm32-microcontrollers-using-cmsis-stmicroelectronics.pdf |
| Application Notes for related Tools & Software | https://www.st.com/resource/en/application_note/an4891-stm32h72x-stm32h73x-and-singlecore-stm32h74x75x-system-architecture-and-performance-stmicroelectronics.pdf |
| Application Notes for related Tools & Software | https://www.st.com/resource/en/application_note/an5001-stm32cube-expansion-package-for-stm32h7-series-mdma-stmicroelectronics.pdf |

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| Application Notes for related Tools & Software | https://www.st.com/resource/en/application_note/an5014-stm32h7x3-smart-power-management-expansion-package-for-stm32cube-stmicroelectronics.pdf |
| Application Notes for related Tools & Software | https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf |
| Application Notes for related Tools & Software | https://www.st.com/resource/en/application_note/an5056-integration-guide-for-the-xcubesbsfu-stm32cube-expansion-package-stmicroelectronics.pdf |
| Application Notes for related Tools & Software | https://www.st.com/resource/en/application_note/an5360-getting-started-with-projects-based-on-the-stm32mp1-series-in-stm32cubeide-stmicroelectronics.pdf |
| Application Notes for related Tools & Software | https://www.st.com/resource/en/application_note/an5361-getting-started-with-projects-based-on-dualcore-stm32h7-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf |
| Application Notes for related Tools & Software | https://www.st.com/resource/en/application_note/an5394-getting-started-with-projects-based-on-the-stm32l5-series-in-stm32cubeide-stmicroelectronics.pdf |
| Application Notes for related Tools & Software | https://www.st.com/resource/en/application_note/an5418-how-to-build-a-simple-usbpd-sink-application-with-stm32cubemx-stmicroelectronics.pdf |
| Application Notes for related Tools & Software | https://www.st.com/resource/en/application_note/an5426-migrating-graphics-middleware-projects-from-stm32cubemx-540-to-stm32cubemx-550-stmicroelectronics.pdf |
| Application Notes for related Tools & Software | https://www.st.com/resource/en/application_note/an5564-getting-started-with-projects-based-on-dualcore-stm32wl-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf |
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